IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 9 December 2024, accepted 24 January 2025, date of publication 4 February 2025, date of current version 12 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3538836

RESEARCH ARTICLE

Graph Multi-Resolution Transformer for Road Traffic Anomaly Detection

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This work was supported by the Korea Institute of Police Technology (KIPoT) through the Korean National Police Agency (KNPA), Development of Integrated Road Traffic Control System and Operation Technology When Autonomous Driving is Mixed With Normal Vehicles under Grant 092021C28S01000.

ABSTRACT Traffic anomaly detection has become increasingly vital for ensuring the safety, efficiency, and resilience of urban transportation systems. The emergence of megacities and the integration of autonomous vehicles (AVs) introduce new challenges for anomaly detection in mixed-traffic environments. AVs exhibit distinct driving behaviors compared to human drivers, resulting in more complex traffic scenarios that necessitate advanced detection methods. This paper presents a novel Graph Multi-Resolution Transformer (GMRT) model designed for traffic anomaly detection in environments where autonomous and human-driven vehicles coexist. Utilizing simulation data from the autonomous driving demonstration zone in South Korea, the GMRT effectively captures temporal patterns at multiple resolutions and leverages graph-based spatial relationships to enhance anomaly detection performance. Unlike traditional models that focus on macro-level traffic flow, our approach operates at a micro-level (lane-based), enabling precise identification of anomalies on a per-lane basis. Comparative experiments against established time-series models demonstrate the superior accuracy and robustness of GMRT, particularly in early anomaly detection across varied temporal horizons. The results underscore the model's potential as a robust solution for contemporary and future urban traffic management challenges, especially in scenarios involving the increasing penetration of AVs.

INDEX TERMS Deep learning, intelligent transportation monitoring, road traffic anomaly detection, multi-resolution learning, time-series analysis.

I. INTRODUCTION

Road traffic anomaly detection has emerged as a critical research area in recent years, owing to its essential role in maintaining the safety, efficiency, and resilience of urban transportation systems [1], [2]. Traffic anomalies, including accidents, unexpected congestion, and abnormal driving behaviors, can significantly disrupt traffic flow, leading to considerable economic losses and increased risks for road users. To mitigate these issues, extensive research has been conducted in traffic anomaly detection, resulting in the development of various statistical, machine learning, and deep learning-based approaches [3], [4].

The associate editor coordinating the review of this manuscript and approving it for publication was Yanli Xu.

The rapid growth of megacities and the advent of autonomous vehicles (AVs) introduce new and intricate challenges for traffic anomaly detection [5]. AVs exhibit driving behaviors distinct from those of human drivers and are anticipated to operate alongside conventional vehicles in mixed-traffic environments. This coexistence generates more complex traffic scenarios, presenting diverse and previously unencountered road conditions. This mixed environment increases the possibility of traffic anomalies due to differences in driving patterns such as following distance, reaction time, and lane-changing behavior. The increased complexity necessitates maintaining optimal traffic flow and minimizing potential confusion or critical accidents, as even minor disruptions can propagate cascading effects throughout the entire road network.

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In this context, the ability to perform rapid and accurate anomaly detection becomes important in mixed-traffic environments involving both autonomous and human-driven vehicles. Early detection and timely responses to traffic anomalies can substantially reduce risks and sustain traffic flow, particularly as the penetration of AVs continues to rise.

This paper introduces a novel traffic anomaly detection method specifically designed for mixed autonomous and human-driven vehicle environments. We simulate traffic scenarios in the autonomous driving demonstration zone in Pangyo, South Korea, serving as a real-world case study. The simulation data from this environment underpins our anomaly detection model. We present a Graph Multi-Resolution Transformer (GMRT) model, which accounts for temporal patterns across various resolutions—such as daily, weekly, and monthly cycles—thereby enhancing anomaly detection performance.

Moreover, while prior studies predominantly utilized deep learning models like Gate Recurrent Units (GRU) [6], Long Short-Term Memory (LSTM) [7], and Temporal Convolutional Networks (TCN) [8], for time-series analysis in traffic anomaly detection, these models often encounter information loss when processing extended time-series data due to difficulties in retaining past temporal features. To overcome this limitation, our model leverages the Transformer architecture [9], renowned for its capability to capture long-range dependencies more effectively, thereby improving detection accuracy for prolonged traffic sequences.

Contrary to existing research that typically focuses on macro-level (link) traffic flow, this paper proposes a micro-level (lane) traffic anomaly detection approach, facilitating precise identification of anomalies on a per-lane basis. This micro-level methodology is increasingly relevant as advanced traffic data collection systems—enabled by advancements in sensor technologies and communication systems—become more prevalent. For instance, smart intersections including edge internet-of-things applications [10], [11] will facilitate the acquisition of lane-specific data, providing a more granular perspective of road conditions.

The key contributions of this paper are as follows.

- *Multi-Resolution Approach*: We propose a multi-resolution methodology for road traffic anomaly detection that accounts for traffic patterns at varying scales, such as daily and weekly cycles, thereby enhancing the detection accuracy of diverse anomaly types.
- *Transformer Architecture Utilization*: We employ the Transformer architecture, recognized for its superior ability to capture long-range dependencies in time-series data, to surmount the limitations of traditional time-series models like LSTM and GRU.
- *Comprehensive Validation*: We validate our approach through experiments using simulated data from mixed autonomous and human-driven vehicle environments. Comparative performance analysis against established time-series models demonstrates the GMRT model's effectiveness and robustness.
- *Micro-Level Detection*: We introduce a micro-level (lane) traffic anomaly detection approach, enabling precise anomaly identification at the lane level, which is critical given the anticipated advancements in traffic data collection systems.

The remainder of this paper is organized as follows. Section II reviews related work on traffic anomaly detection and multi-resolution learning. Section III defines the problem and details the proposed GMRT model. Section IV describes the data utilized and the experimental procedures. Section V presents the experimental results, and Section VI concludes the study.

II. RELATED WORK

This section examines related research from two perspectives: road traffic anomaly detection and multi-resolution analysis.

A. ROAD TRAFFIC ANOMALY DETECTION

Early investigations into road traffic anomaly detection primarily employed statistical models. Techniques such as Gaussian Mixture Models (GMMs) [2] and Principal Component Analysis (PCA) [12] were utilized to model and detect traffic anomalies. While these methods proved effective in simple traffic scenarios, they struggled to manage the escalating complexity of modern road networks, especially with the proliferation of real-time, high-dimensional traffic data.

The advent of machine learning introduced more sophisticated approaches to anomaly detection in traffic systems. Supervised learning models, including Support Vector Machines (SVMs), Random Forests [13], and k-Nearest Neighbors (k-NN) [14], have been extensively adopted in this domain. These models rely on labeled datasets containing both normal and anomalous data for training. However, the scarcity of labeled data in large-scale traffic networks significantly limits their applicability. Consequently, unsupervised learning techniques, particularly clustering algorithms like k-means and DBSCAN, have gained popularity. These methods can detect anomalies without requiring labeled data by identifying outliers that deviate from the learned distribution of normal traffic patterns.

Recent advancements in deep learning have further expanded the capabilities of traffic anomaly detection systems. Models such as GRU [6] and LSTM networks [7] have been employed to capture temporal dependencies in traffic data. Additionally, Convolutional Neural Networks (CNNs) have been explored for traffic anomaly detection [8], particularly when analyzing traffic data from sensor networks or CCTV cameras, as CNNs excel in extracting spatial features crucial for understanding traffic dynamics.

Given the inherently spatial structure of road networks, graph-based approaches have gained attraction in recent

years. These methods represent the road network as a graph, where nodes correspond to road segments or intersections and edges denote the connections between them. Techniques such as Graph Convolutional Networks (GCNs) [15], graph autoencoders [8], and graph attention mechanisms [6] have been employed to capture spatial dependencies across the network. These models facilitate anomaly detection by analyzing how disruptions in one part of the network affect connected road segments.

Despite the successes of these approaches in enhancing road traffic anomaly detection, they often overlook the challenges associated with long-length time series. To address this gap, our proposed model is based on the Transformer architecture, which has demonstrated significant capabilities in modeling long-range temporal dependencies, making it well-suited for long time-series analysis in traffic anomaly detection.

B. MULTI-RESOLUTION LEARNING

The concept of multi-resolution features has been extensively studied in the field of image analysis [16], [17], [18]. In this domain, multi-resolution methods enable the extraction of different frequency characteristics from objects, allowing models to capture both coarse and fine details simultaneously. These methods process input images at multiple scales, enhancing the model's ability to learn high-level features, such as the location of objects within a larger context (e.g., the brain in medical imaging), while still preserving detailed local structures. This approach offers several advantages, including improved feature representation, robustness to noise, and better generalization across diverse image patterns.

Building on these successes in image analysis, multi-resolution techniques have also been applied to time-series data [19], [20], [21], [22]. Traditionally, wavelet decomposition has been used for multi-resolution analysis in time-series [23]. It allows the decomposition of a signal into different frequency components, enabling models to capture both short- and long-term temporal dependencies, which are critical for tasks involving complex temporal patterns.

In recent years, several advanced models, such as TimesNet [19], Pyraformer [21], Scaleformer [20], and MTST [24] (Multi-resolution Time-Series Transformer), have introduced sophisticated architectures designed to handle multi-resolution features for time-series forecasting. These models leverage multi-resolution analysis to enhance forecasting accuracy by capturing temporal patterns at different granularities. For instance, TimesNet transforms one-dimensional time-series data into two-dimensional tensors, enabling the model to better capture and represent temporal complexity. Pyraformer models temporal dependencies of varying lengths using pyramidal attention mechanisms, while Scaleformer employs shared weights and an innovative normalization process to refine predictions at multiple scales.

However, despite these advances, most prior studies have focused on applying multi-resolution analysis techniques to time-series forecasting, rather than anomaly detection. In this study, we bridge this gap by applying multi-resolution analysis to the context of anomaly detection in traffic networks. Our approach leverages the strengths of multi-resolution features to detect anomalies that might otherwise be missed by single-resolution models.

III. METHOD

A. PROBLEM STATEMENT

We consider a road network π, comprising M links, each containing a varying number of lanes. The i-th lane of the k-th link is denoted as v_i^k. Our objective is to detect anomalies in the k-th link, v^k, using the traffic attributes from its constituent lanes $\{v_i^k | i \in \mathbb{N}, i > 0\}$.

The anomaly detection outcome for the link v^k at time t is determined by the anomaly scores of lanes v_i^k at t and a detection threshold θ as

$$y_t^k = \begin{cases} 0, & if\ \forall score_{i,t}^k < \theta \\ 1, & otherwise, \end{cases} \tag{1}$$

where y_t^k represents the anomaly label of link v^k at time t, and $score_t^k$ is the anomaly score of v^k at time t. The label y_t^k is 0 if the anomaly scores for all lanes in link v^k are below θ; otherwise, the label is 1, indicating that link v^k is anomalous at time t.

Historical observations of traffic attributes at time t are denoted as $x_{t-T+1:t} \in \mathbb{R}^{T \times N \times C}$, where N is the total number of lanes in the road network π, C is the number of the traffic attributes, and T is the length of the time slice. The anomaly score, $score_{i,t}^k$, is obtained using the historical time slice $x_{t-T+1:t}$ as input to the anomaly detection model f, and a scoring function g as shown as

$$score_{i,t}^k = \mathrm{g}\left(i, k, f\left(x_{t-T+1:t}\right)\right). \tag{2}$$

In this paper, the anomaly detection model f is a reconstruction-based model. Therefore, the scoring function g should be capable of measuring the discrepancies between the original input $x_{t-T+1:t}$ and the reconstructed output $\hat{x}_{t-T+1:t}$. Various functions can be employed for g, such as root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE) or Huber loss.

If RMSE is utilized as the scoring function g, the anomaly score $score_{i,t}^k$ is calculated as

$$\mathrm{g}\left(i, k, f\left(x_{t-T+1:t}\right)\right) = \sqrt{\frac{1}{C}\sum_{j=1}^{C}\left(x_{i,j,t-T+1:t}^k - \hat{x}_{i,j,t-T+1:t}^k\right)^2}, \tag{3}$$

where $x_{i,j,t-T+1:t}^k$ represents the j-th attribute values of the i-th lane of the k-th link from time t-T + 1 to t. For

clarity, all notations used in this study are summarized in Table 1.

B. GRAPH MULTI-RESOLUTION TRANSFORMER

The architecture of GMRT is inspired by MTST which used multi-resolution temporal features to forecast long-term time-series. GMRT, at first, transforms the input traffic attributes into graph-structured data to capture the spatial relationships among traffic links. Moreover, to incorporate cyclical traffic patterns, time information is added using a time encoding process.

Second, the graph-structured data passes through L layers, each comprising multiple branches for extracting multi-resolution temporal features. Within each layer, the input data is tokenized into several patches, processed through a Transformer encoder with multi-head self-attention mechanisms. Finally, the outputs from each branch are flattened, concatenated, and fused by a linear projection layer.

TABLE 1. Summary of notations.

Symbol	Description
π	Traffic network
v_i^k	The i-th lane of the k-th link
y_t^k	The anomaly label of the k-th link at time t
N	The total number of lanes
M	The total number of links
C	The number of traffic attributes
T	Length of look-back window
$x_{i,j,t-T+1:t}^k$	The j-th traffic attribute values of the i-th lane of the k-th link from time t-T+1 to t
A	Adjacency matrix
$e_{p\to q}$	Indicator of whether link v^p has a road leading to link v^q
$\tilde{x}_{t-T+1:t}$	Graph-structured traffic attributes at time t
p_b	Patch size of branch b
s_b	Stride size of branch b
$\boldsymbol{Y}^l$	Output of the (l-1)-th layer in GMRT
$\boldsymbol{y}^l_{1+r\times s_b:p_b+r\times s_b}$	The r-th patch in $\boldsymbol{Y}^l$, which has p_b length and s_b stride size

1) GRAPH REPRESENTATION OF TRAFFIC NETWORK

To capture the spatial relationships among different links, the raw input data is transformed into a graph structure using an adjacency matrix A. The adjacency matrix $A \in \mathbb{R}^{N\times N}$ is constructed based on road connectivity. If link v^p has a road to link v^q, the element value of A, A_{pq} is 1; otherwise 0 as

$$A_{pq} = \begin{cases} 1, \text{ if } p \neq q \text{ and } e_{p\to q} = \text{True} \\ 0, \text{ otherwise.} \end{cases} \tag{4}$$

Then, each raw input attributes at every time step, $x_{t'} \in \mathbb{R}^{N\times C}$, is multiplied by the adjacency matrix A. The resulting values are concatenated along the time dimension T as

$$\tilde{x}_{t-T+1:t} = [A \cdot x_{t-T+1} \parallel A \cdot x_{t-T+2} \parallel \ldots \parallel A \cdot x_t], \tag{5}$$

where $\tilde{x}_{t-T+1:t}$ represents the graph-structured traffic attributes at time t, and $\parallel$ denotes the concatenation operation.

2) TIME ENCODING

Time information is crucial for capturing the traffic dynamics and dependencies over time, particularly due to the cyclical nature of traffic patterns influenced by factors such as rush hours. Therefore, in this paper, the 24-hour day is divided into five intervals: early morning (00:00 $\sim$ 08:00), morning rush hour (08:00 $\sim$ 10:00), afternoon (10:00 $\sim$ 18:00), evening rush hour (18:00 $\sim$ 20:00), and night (20:00 $\sim$ 24:00). Each time interval is encoded using one-hot encoding, resulting in binary attributes (0 or 1). Then they are concatenated with the graph-structured traffic attributes $\tilde{x}_{t-T+1:t}$.

3) MULTI-RESOLUTION LEARNING

Previous studies using the Transformer architecture for time-series analysis treated each time stamp as a token (point-wise tokenization) for the transformer's input [25], [26]. However, recent studies have shown that the point-wise tokenization is hard to extract local semantic information using a single time point [27]. Instead, aggregating several time stamps into patches facilitates the model in learning local semantic representations (patch-based tokenization) [28].

The patch-based tokenization is applied to each attribute (channel) in data. This means that input tokens contain only single-attribute information without channel-mixing. The patches have different sizes, creating multiple branches for multi-resolution learning. Larger patches generate coarser temporal feature maps capable of detecting long-cyclical traffic patterns in a long-time range, while smaller patches capture finer temporal features suitable for identifying instantaneous events within shorter time frames.

Additionally, patch-tokenization is performed at each layer of the model to enhance the learning of rich multi-resolutional features. By repetitively dividing and combining the input across multiple layers, the model effectively captures a diverse array of temporal features at various resolutions. By continuously refining feature representations at each layer, the model enhances its ability to detect both long-term and short-term anomalies, addressing the complexities inherent in mixed autonomous and human-driven vehicle environments. This multi-layered patch-based approach enables the Transformer to adeptly handle complex and dynamic traffic patterns.

The patch-based tokenization process is formulated as follows. Here, the notation on the attribute is ignored for formula simplicity. The output of the $(l-1)$-th layer $\boldsymbol{Y}^{l-1} \in \mathbb{R}^{T\times N}$ is given as input of the l-th layer, and in the l-th layer, $\boldsymbol{Y}^{l-1}$ is divided into different length of patches in B branches as

$$\boldsymbol{Y}_b^{l-1} = \left\langle \boldsymbol{y}^{l-1}_{1:p_b}, \boldsymbol{y}^{l-1}_{1+s_b:p_b+s_b}, \boldsymbol{y}^{l-1}_{1+2s_b:p_b+2s_b}, \ldots, \boldsymbol{y}^{l-1}_{d_{l-1}-p_b:d_{l-1}} \right\rangle, \tag{6}$$

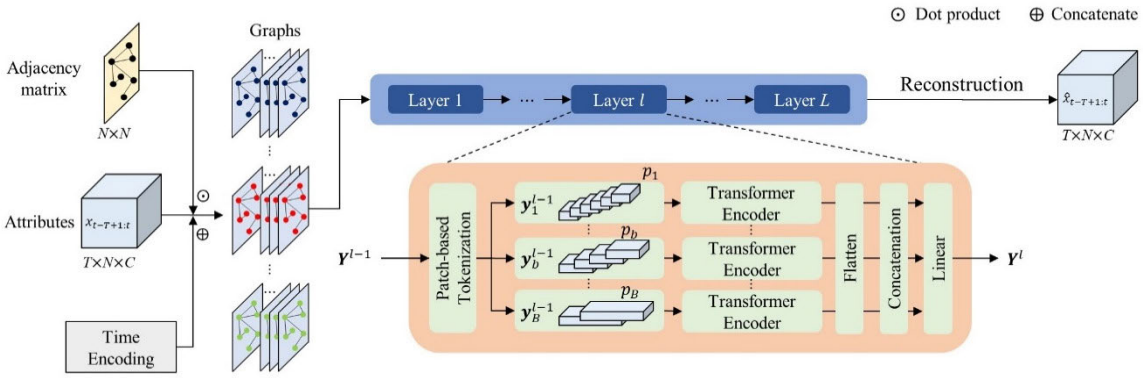


FIGURE 1. The architecture of graph multi-resolution transformer (GMRT).

where p_b is the patch size of the b-th branch, and s_b is the stride size of the b-th branch.

4) TRANSFORMER ENCODER

The Transformer encoder captures complex temporal dependencies and interactions within the traffic data. Utilizing the tokenized patches generated in each branch, the Transformer encoder employs a self-attention mechanism to model both short-term and long-term dependencies effectively.

In the Transformer encoder, the multi-head self-attention mechanism which allows the model to weigh the significance of different tokens relative to each other is calculated. The attention score between the r-th and s-th token in branch b, α_{rs}, is computed as

$$\begin{aligned}\alpha_{rs} &= softmax(\frac{\mathbf{Q}_r^b \left(\mathbf{K}_s^b\right)^{\mathrm{T}}}{\sqrt{d_k}})\\ \mathbf{Q}_r^b &= \boldsymbol{W}_{\mathrm{Q}} \cdot \boldsymbol{y}_{1+r\times s_b:p_b+r\times s_b}^{l-1} + \mathbf{b}_{\mathrm{Q}}\\ \mathbf{K}_s^b &= \boldsymbol{W}_{\mathrm{K}} \cdot \boldsymbol{y}_{1+s\times s_b:p_b+s\times s_b}^{l-1} + \mathbf{b}_{\mathrm{K}},\end{aligned} \tag{7}$$

where $\boldsymbol{W}_{\mathrm{Q}}$ and $\boldsymbol{W}_{\mathrm{K}}$ are the learnable weight matrices for queries and keys, respectively, and $\mathbf{b}_{\mathrm{Q}}$ and $\mathbf{b}_{\mathrm{K}} \in \mathbb{R}^{d\times d_k}$ are the bias terms. d_k represents the dimensionality of the queries and keys.

Then, the output of the self-attention mechanism for the r-th token in branch b, $\tilde{\boldsymbol{y}}_{1+r\times s_b:p_b+r\times s_b}^{l-1}$, is calculated as

$$\begin{aligned}\tilde{\boldsymbol{y}}_{1+r\times s_b:p_b+r\times s_b}^{l-1} &= \alpha_{rs}\mathbf{V}_s^b\\ \mathbf{V}_s^b &= \boldsymbol{W}_{\mathrm{V}} \cdot \boldsymbol{y}_{1+s\times s_b:p_b+s\times s_b}^{l-1} + \mathbf{b}_{\mathrm{V}},\end{aligned} \tag{8}$$

where $\boldsymbol{W}_{\mathrm{V}}$ is the learnable weight matrix for values, and $\mathbf{b}_{\mathrm{V}}$ is the bias term.

Following the self-attention mechanism, residual connections, normalization, and feed-forward networks are applied as the original Transformer encoder. Finally, the final outputs from each branch are flattened, concatenated, and fused using a linear layer as

$$\boldsymbol{Y}^l = \boldsymbol{W}^{l-1}\left[\boldsymbol{o}_1^{l-1} \parallel \ldots \parallel \boldsymbol{o}_2^{l-1} \parallel \ldots \parallel \boldsymbol{o}_B^{l-1}\right] + \boldsymbol{b}^{l-1} \tag{9}$$

where $\boldsymbol{Y}^l$ is the output of the l-th layer, $\boldsymbol{W}^{l-1} \in \mathbb{R}^{B\times d_{l-1}}$ and $\boldsymbol{b}^{l-1} \in \mathbb{R}^{1\times d_{l-1}}$ are the weight and bias matrices for linear projection in the $(l-1)$-th layer, and $\boldsymbol{o}_b^{l-1} \in \mathbb{R}^{1\times d_{l-1}}$ represents the output values of the Transformer encoder.

C. LOSS FUNCTION AND ANOMALY SCORE

Our model adopts a reconstruction-based approach for anomaly detection. Consequently, the loss function during training aims to minimize the discrepancy between the original input $x_{i,j,t-T+1:t}^k$ and the reconstructed output $\hat{x}_{i,j,t-T+1:t}^k$. We employ the MSE as the loss function as

$$\text{Loss} = \frac{1}{NCM}\sum_{i=1}^{N}\sum_{j=1}^{C}\sum_{k=1}^{M}\left(x_{i,j,t-T+1:t}^k - \hat{x}_{i,j,t-T+1:t}^k\right)^2. \tag{10}$$

The model is trained using an unsupervised learning approach, which utilizes only normal data for training and validation. After training the model, test data that include anomalies is used to evaluate the detection performance of the model.

IV. EXPERIMENTS

A. DATA

The dataset used in this study is constructed using VISSIM, a microscopic traffic simulation software due to inherent challenges of acquiring large-scale, labeled traffic anomaly data from real-world environments. The simulation replicates traffic environments where autonomous vehicles and conventional vehicles coexist within the autonomous driving demonstration zone in Pangyo, South Korea. The simulation environment meticulously considers actual physical conditions, including the number of lanes, lane widths, slope degrees, and traffic signal phase systems.

Fig. 2 shows the constructed traffic network. It has 13 intersections, 66 links, and 256 lanes. The number of vehicles and their driving directions are based on actual traffic records from the network. Over 24-hour period,

the ratio of conventional vehicles to autonomous vehicles is maintained at 7:3, and the autonomous vehicles are configured to operate at level 4 autonomy (full automation) [29] by adjusting the parameters of the Wiedemann 99 model within VISSIM. Speed limitations of vehicles are set to 50 km/h on normal roads and 30 km/h on alleys.



FIGURE 2. The actual traffic network (a) and the simulation network (b) of Pangyo, South Korea.

The simulation time spans 14 days comprising 7 days without anomalies and 7 days with anomalies. The first 7 days without anomalies serves as normal traffic condition, and the second 7 days with anomalies is used to evaluate the detection model. Traffic anomalies are simulated by blocking lanes through the creation of parking lots or reducing vehicle speeds.

In 2023, traffic anomalies in Gyeonggi-do, South Korea, primarily consisted of car collisions (57%), vehicle breakdowns (32%), fallen objects on roads (6%), and road maintenance activities (3%) [30]. These incidents frequently cause significant traffic congestion. To realistically simulate such conditions based on these statistics, we designed three types of traffic anomalies involving lane blockages (AN1, AN2, and AN3), each representing varying levels of traffic disruption. Additionally, adverse weather conditions are simulated by reducing vehicle speeds to reflect weather-induced slowdowns (AN4).

To represent diverse types of anomalies, different severities are implemented as detailed in Table 2. Additionally, anomalies occur at different times and have varying durations (5, 10, 15, 20, 60 minutes). The anomalies are generated in 15 links which are randomly selected. The generated anomalies for 7 days are described in Table 3.

Five traffic attributes are collected for each lane every 5 minutes: the total number of passing vehicles, harmonic mean speed of all vehicles, arithmetic mean speed of all vehicles, total delay time of all vehicles, and occupancy rate. The used traffic attributes are detailed in Table 4.

B. EXPERIMENTAL SETUP

The dataset is divided into three parts: 5 days for training (day 1 – day 5), 2 days for validation (day 6 – day 7), and 7 days for testing (day 8 – day 14). The validation data is used to compute validation loss every epoch, and the training is early stopped when the validation loss does not decrease over several consecutive epochs to prevent overfitting. The test data is continuous with the validation data, ensuring consistent environmental conditions between training and testing phases. Also, the anomaly detection threshold θ is established based on the maximum anomaly score observed in the validation set.

TABLE 2. Generated traffic anomalies.

Anomaly event	Simulation method	Examples in real world
AN1	Blocking 10% of a lane	minor collision, car breakdown, small/single fallen object
AN2	Blocking 20% of two lanes	serious collision, large/multiple fallen object(s)
AN3	Blocking 10% of the last right lane	road maintenance
AN4	Reducing vehicle speed by 20%	bad weather conditions

Optimal hyperparameters of the GMRT model are searched using a grid-search method with the following parameters: the number of layers = {1, 2, 3}, the number of attention heads = {4, 8, 16, 32}, the number of patches = {6, 12, 36, 144, 216}. The stride size is set to half of the corresponding patch size. Both the model dimension and the fully connected layer dimension are fixed at 128. A dropout rate of 0.1 is applied, and the learning rate is initialized at 0.001 and adjusted using a plateau-based scheduling mechanism, which reduced the learning rate if no improvement in validation performance is observed.

C. EVALUATION

The performance of the proposed GMRT model is evaluated against several baseline models, LSTM, GRU, Transformer, Informer, and Autoformer. LSTM and GRU are the well-established model for time-series data, while Transformer, Informer, and Autoformer are advanced architectures designed to handle long time-series dependencies.

The performance of the model is calculated by the anomaly event not using the time points because it has no meaning to measure the performance at a time point.

As a performance metric, we use the recall, which measures the proportion of actual anomalies that are correctly identified by the model as

$$\text{Recall} = \frac{\text{True Positive}}{\text{True positive} + \text{False Positive}}. \tag{11}$$

Due to the interconnected nature of traffic networks—where anomalies in one link can propagate and influence other links—recall has been widely adopted as the primary

TABLE 3. Generated anomaly events across different days and time slots.

Link #	Day 8	Day 9	Day 10	Day 11	Day 12	Day 13	Day 14
1		AN2 (10:00-10:20)	AN4 (06:00-07:00)		AN1 (13:00-13:35)		
2			AN4 (06:00-07:00)				
3			AN4 (06:00-07:00)	AN1 (20:30-20:40)		AN3 (14:00-14:20)	
4			AN4 (06:00-07:00)				AN3 (15:30-15:40)
5	AN3 (09:00-09:05)		AN4 (06:00-07:00)		AN1 (09:30-09:40)		
6		AN3 (18:30-19:00)	AN4 (06:00-07:00)				
7			AN4 (06:00-07:00)	AN1 (15:30-15:55)			
8			AN4 (06:00-07:00)			AN2 (17:00-17:15)	
9			AN4 (06:00-07:00)	AN2 (07:00-07:30)			
10	AN1 (14:00-14:30)		AN4 (06:00-07:00)				
11			AN4 (06:00-07:00)		AN3 (21:00-21:15)		AN1 (06:00-06:20)
12		AN1 (22:00-22:20)	AN4 (06:00-07:00)				
13			AN4 (06:00-07:00)				AN2 (16:00-16:20)
14	AN1 (18:00-18:15)		AN4 (06:00-07:00)				
15			AN4 (06:00-07:00)			AN1 (12:00-12:30)	

TABLE 4. Traffic attributes.

Attributes	Unit	Range
The total number of passing vehicles	-	[0, ∞]
Harmonic/arithmetic mean of the speed of all vehicles	km/h	[0, ∞]
Total delay time of all vehicles	second	[0, ∞]
Occupancy rate	%	[0, 100]

performance metric in evaluating traffic anomaly detection models in previous studies. Note that precision is not reported because the complexity of traffic networks can result in unidentified anomalies, and the anomalies in neighboring links increase the anomaly score of the tested link.

To evaluate the model's capability in early anomaly detection, performance is measured across various detection horizons: on-time, +30 minutes, +60 minutes, +90 minutes, and +120 minutes. "On-time" detection refers to anomalies being identified immediately as they occur, while "+n minutes" indicates that anomalies are detected within n minutes after the event occurs.

V. RESULTS

The detection performance is presented as the mean recall score along with the standard deviation of the tested links results as shown in Table 5.

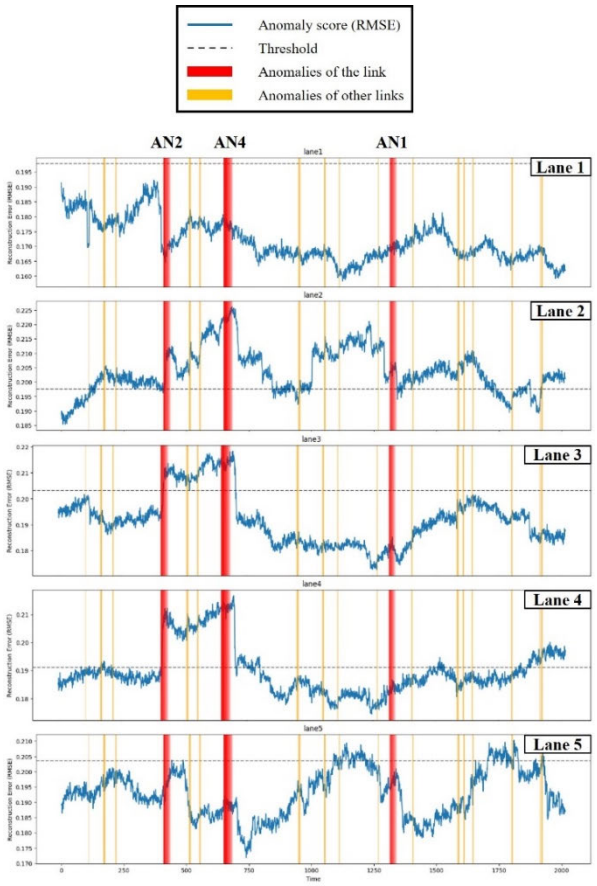


FIGURE 3. Anomaly scores across lanes in link 1.

A. TRAFFIC ANOMALY DETECTION

The GMRT model consistently outperforms all baseline models across varying values of T (36, 72, 144, 288), demonstrating its robustness and effectiveness in traffic anomaly detection. At $T = 36$, GMRT achieves an average recall of 0.5293, surpassing LSTM (0.4488), GRU (0.4488), Transformer (0.4773), Informer (0.5000), and Autoformer (0.4911). At $T = 72$, GMRT records an average recall of 0.6369, outperforming LSTM (0.4555), GRU (0.5666), Transformer (0.4355), Informer (0.5488), and Autoformer (0.6488). At $T = 144$, GMRT attains an average recall of 0.7089, exceeding LSTM (0.4533), GRU (0.5911), Transformer (0.5622), Informer (0.5666), and Autoformer (0.7011). At $T = 288$, GMRT achieves the highest average recall of 0.7755, outperforming LSTM (0.6651), GRU (0.6844), Transformer (0.6775), Informer (0.6826), and Autoformer (0.7222).

Across all values of T, GMRT maintains superior recall scores compared to baseline models, underscoring its efficacy in accurately detecting traffic anomalies. The integration of multi-resolution temporal analysis and graph-based spatial modeling in GMRT enables it to capture complex and dynamic traffic patterns more effectively than models relying solely on temporal or spatial features.

TABLE 5. Road traffic anomaly detection results (recall).

Models	T	On time	+30 min.	+60 min.	+90 min.	+120 min.	Avg.
LSTM	36	0.4222±0.3861	0.4444±0.3244	0.4444±0.3288	0.4667±0.3408	0.4667±0.3285	0.4488±0.3417
	72	0.3889±0.3792	0.4556±0.3582	0.4556±0.3366	0.4889±0.3280	0.4889±0.3203	0.4555±0.3444
	144	0.3667±0.3821	0.4333±0.3288	0.4889±0.3283	0.4889±0.3208	0.4889±0.3128	0.4533±0.3345
	288	0.6478±0.3191	0.6444±0.3203	0.6778±0.3191	0.6778±0.3191	0.6778±0.3191	0.6651±0.3193
GRU	36	0.4889±0.4582	0.4444±0.3401	0.4444±0.3305	0.4333±0.3271	0.4333±0.2400	0.4488±0.3391
	72	0.5222±0.3950	0.5444±0.3291	0.5667±0.3282	0.6000±0.3309	0.6000±0.2864	0.5666±0.3339
	144	0.5556±0.4180	0.6000±0.3382	0.6000±0.3194	0.6000±0.3288	0.6000±0.2803	0.5911±0.3369
	288	0.6444±0.4027	0.6444±0.3325	0.7000±0.3285	0.7000±0.3285	0.7333±0.2731	0.6844±0.3330
Transformer	36	0.4589±0.3604	**0.4778**±0.3406	0.4778±0.3604	0.4778±0.2638	0.4944±0.2509	0.4773±0.3152
	72	0.4444±0.3325	0.4333±0.3280	0.4333±0.3155	0.4333±0.2513	0.4333±0.2604	0.4355±0.2975
	144	0.5111±0.3408	0.5444±0.3155	0.5444±0.3208	0.5889±0.2812	0.6222±0.2400	0.5622±0.2996
	288	0.6589±0.3325	0.6778±0.3115	0.6733±0.3261	0.6889±0.2395	0.6889±0.2395	0.6775±0.2898
Informer	36	0.4556±0.4680	0.4556±0.4387	0.4889±0.4400	0.5111±0.4601	**0.5889**±0.4283	0.5000±0.4470
	72	0.4889±0.4561	0.5444±0.4222	0.5444±0.4381	0.5333±0.4282	0.6333±0.4190	0.5488±0.4327
	144	0.5222±0.4428	0.5444±0.4209	0.5667±0.4366	0.6000±0.4355	0.6000±0.4157	0.5666±0.4303
	288	0.6222±0.4448	0.6689±0.4247	0.6889±0.4247	0.6889±0.4247	0.7444±0.4247	0.6826±0.4287
Autoformer	36	0.4667±0.3869	0.4333±0.3642	0.5000±0.3762	0.5000±0.3612	0.5556±0.3680	0.4911±0.3713
	72	0.5444±0.3702	0.5889±0.3608	**0.6222**±0.3844	**0.7444**±0.3802	**0.7444**±0.3678	**0.6488**±0.3726
	144	0.6000±0.3811	0.6667±0.3488	**0.6833**±0.3875	0.7667±0.3744	0.7889±0.3608	0.7011±0.3705
	288	0.6778±0.4027	**0.7222**±0.3644	0.7222±0.3644	0.7444±0.3556	0.7444±0.3203	0.7222±0.3614
GMRT	36	**0.4912**±0.3642	0.4667±0.3522	**0.5333**±0.3892	**0.5667**±0.2904	**0.5889**±0.3283	**0.5293**±0.3448
	72	**0.5889**±0.3682	**0.5956**±0.3438	0.6111±0.3011	0.6667±0.2755	0.7222±0.3366	0.6369±0.3250
	144	**0.6222**±0.3544	**0.6689**±0.3422	0.6756±0.3072	**0.7778**±0.2945	**0.8000**±0.3059	**0.7089**±0.3208
	288	**0.7111**±0.3477	**0.7222**±0.3128	**0.7778**±0.2931	**0.8333**±0.2591	**0.8333**±0.2597	**0.7755**±0.2944

Moreover, GMRT's recall performance improves consistently as T increases,indicating its ability to effectively leverage longer temporal dependencies for enhanced anomaly detection. While baseline models such as LSTM and GRU show moderate performance at lower T values, their recall scores do not scale as effectively with increasing T, unlike GMRT. Transformer-based models exhibit fluctuating performance, with Autoformer approaching GMRT's performance at higher T values but still trailing slightly.

Overall, the experimental results underscore the effectiveness of the proposed GMRT model in achieving higher detection accuracy, especially in early anomaly detection. The incorporation of graph-structured inputs and multi-resolution learning allows the GMRT model to capture complex, interconnected traffic dynamics, making it more robust and reliable for real-world traffic anomaly detection tasks.

B. CASE STUDY

To further illustrate the effectiveness of the GMRT model in detecting traffic anomalies, we present a detailed case study visualized in Fig. 3. This figure shows the anomaly scores across five lanes of the link 1 in the test data.

In each sub-plot, the blue line represents the anomaly score for each lane, while the dashed line indicates the anomaly detection threshold. Red vertical bars highlight the generated anomalies on the respective lanes, while the yellow bars indicate anomalies on other links in the network.

The figure shows the anomaly scores surpass the threshold, corresponding to specific traffic events. For example, in the first anomaly event (AN2), lane blocking was generated in both the lane 2 and lane 3. The anomaly scores exceeding the threshold are observed in the lane 2 and 3. Additionally, in lane 4, the anomaly score also exceeds the threshold. This result demonstrates that an anomaly event in one lane can affect the traffic flow in adjacent lanes, leading to elevated anomaly scores across the connected lanes. This finding emphasizes the interconnected nature of traffic dynamics, where anomalies in one lane can influence the traffic patterns in neighboring lanes, and it highlights the importance of considering spatial relationships when detecting traffic anomalies.

In the second anomaly event (AN4), the generated anomaly event was a reduction in speed of vehicles in all links. As expected, the anomaly scores of the lane 2, 3, and 4 are very high and exceeds the threshold. Conversely, lane 1 is designated for left-turning vehicles where traffic density is typically lower. This is why the anomaly scores maintains below the threshold.

In the third anomaly event (AN1), a lane blocking was created in lane 2 in the simulation. As expected, the anomaly score for the lane 2 rises above the threshold, successfully detecting the traffic disruption in that lane.

This case study shows that the GMRT model effectively detects traffic anomalies by utilizing both spatial and temporal information. The anomaly score increases at key anomaly points, and the propagation of these anomalies across lanes underscores the model's ability to capture the spatial dependencies inherent in road networks.

VI. CONCLUSION

This paper introduced a novel Graph Multi-Resolution Transformer (GMRT) model designed to detect road traffic anomalies in mixed environments where autonomous vehicles and human-driven vehicles coexist. The proposed GMRT model demonstrates significant performances over existing methods by leveraging both multi-resolution temporal features and graph-based spatial relationships. Through comprehensive experiments on simulated data from the autonomous driving demonstration zone in South Korea, the GMRT model consistently outperformed traditional time-series models such as LSTM, GRU, Transformer, Informer, and Autoformer across various detection horizons.

The key advantages of the GMRT model include its ability to capture traffic anomalies at a micro-level (lane-specific) and its robustness in early anomaly detection, which is critical for managing increasingly complex urban traffic networks. The model's performance improves with longer temporal dependencies, highlighting its scalability and effectiveness in real-world traffic scenarios involving autonomous vehicles.

Overall, the GMRT model presents a promising solution for the next generation of intelligent transportation systems, particularly as autonomous vehicles penetration continues to grow. Future work may extend the model's application to real-world traffic data and explore its performance in different urban contexts.

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